Technology Stack

AI-Based Threat Intelligence Platform

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Components s Technologies Used

S.No	Component	Description	Technology (Used)
1	User Interface	Analyst dashboard, search, KPIs, reports	**Streamlit**, Plotly
2	Data Collection	Fetching threat data from external API	**AlienVault OTX API** (via Python `requests`)
3	Database	Store collected threat indicators	**SQLite**
4	Data Preprocessing	Normalization C cleaning	**Pandas**
5	Data Analysis	Frequency counts, distribution, anomaly detection	**Pandas**, **scikit-learn (IsolationForest)**
6	Visualization	Charts C maps for threats and trends	**Plotly Express**
7	Export	Download processed data	CSV export using Pandas

Application Characteristics

S.No	Characteristic	Implementation
1	Open-Source Frameworks	Streamlit, Pandas, Plotly, scikit-learn
2	Security	API key stored via environment variable (OTX API)
3	Data Scope	Threat feeds limited to AlienVault OTX (100 indicators per request)
4	Scalability	Prototype scale (local database, single API feed)
5	Features	Date range filter, KPIs, anomaly detection toggle, search, CSV export